

Homework Assignment 3 - Coding Part Write-up

Networks and Markets

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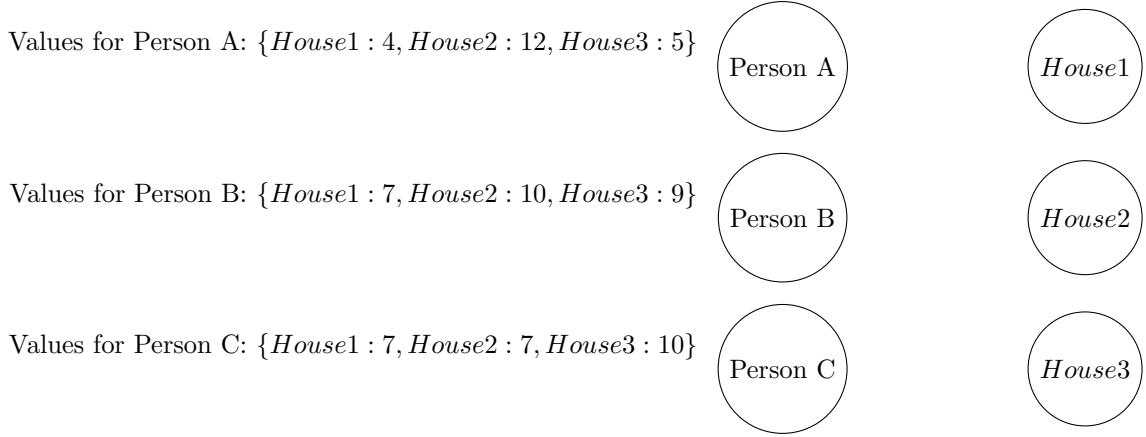
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Part 4: Implementing Matching Market Pricing

1 Question 7

(b) Consider the matching market example in Lecture 5 Page 7:

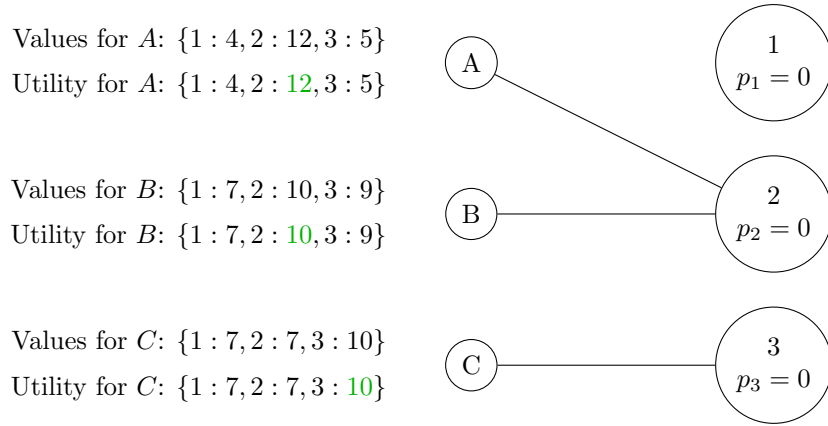


Formally, the matching market context is $\Gamma = (\{A, B, C\}, \{1, 2, 3\}, v)$, where v is the valuation function defined as follows:

$$\begin{aligned}v_A(1) &= 4, v_A(2) = 12, v_A(3) = 5 \\v_B(1) &= 7, v_B(2) = 10, v_B(3) = 9 \\v_C(1) &= 7, v_C(2) = 7, v_C(3) = 10\end{aligned}$$

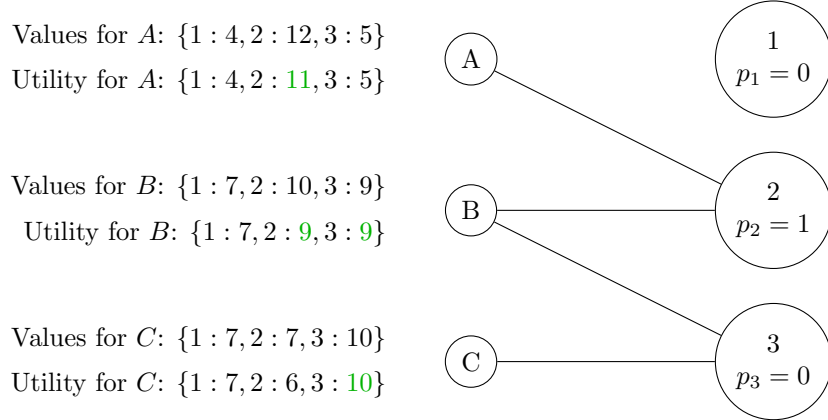
We turn to run the algorithm of Theorem 8.8 to find a market equilibrium (p, M) to find the maximum social value, in order to validate our implementation's output. We begin by initializing the prices vector $\vec{p} \equiv 0$ to be the zero vector. We then proceed to run the algorithm, updating the prices vector until there is a perfect matching M in the induced preferred choice graph for $(\Gamma, \vec{p})$:

1. Observing the following *induced preferred-choice graph* from $(\Gamma, \vec{p})$:



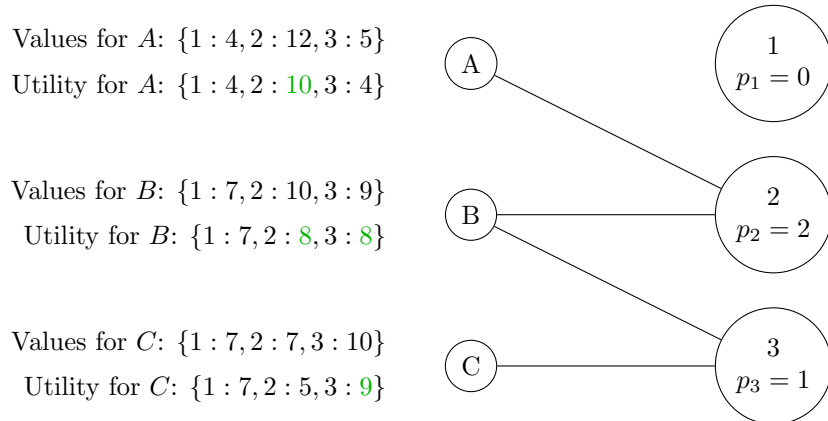
There obviously isn't a perfect matching as $S = \{A, B\}$ is a constricted set with $|N(S)| = |\{2\}| = 1 < 2 = |S|$ (which, by a theorem we've seen in class, implies that there isn't a perfect matching). Thus, we raise the prices for all items in $N(S)$ by 1, and update the prices vector $\vec{p}$ accordingly. The updated prices vector is $\vec{p} = (a : 0, b : 1, c : 0)$. Not all prices are greater than zero, so we don't perform a shift operation, and we proceed to the next iteration.

2. Observing the following *induced preferred-choice graph* from $(\Gamma, \vec{p})$:



There obviously isn't a perfect matching as $S = \{A, B, C\}$ is a constricted set with $|N(S)| = |\{2, 3\}| = 2 < 3 = |S|$ (which, by a theorem we've seen in class, implies that there isn't a perfect matching). Thus, we raise the prices for all items in $N(S)$ by 1, and update the prices vector $\vec{p}$ accordingly. The updated prices vector is $\vec{p} = (a : 0, b : 2, c : 1)$. Not all prices are greater than zero, so we don't perform a shift operation, and we proceed to the next iteration.

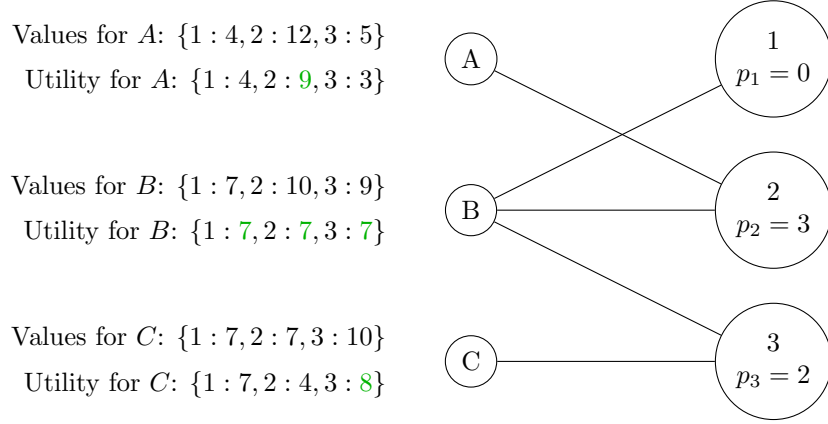
3. Observing the following *induced preferred-choice graph* from $(\Gamma, \vec{p})$:



Similar to the previous iteration, we raise the prices for $\{2, 3\}$, and update the prices vector

$\vec{p}$ accordingly. The updated prices vector is $\vec{p} = (a : 0, b : 3, c : 2)$. Not all prices are greater than zero, so we don't perform a shift operation, and we proceed to the next iteration.

4. Observing the following *induced preferred-choice graph* from $(\Gamma, \vec{p})$:



And there is a perfect matching in the induced preferred choice graph, which is $M = \{\{A, 2\}, \{B, 1\}, \{C, 3\}\}$. Thus, the market equilibrium is $(\vec{p}, M) = ((1 : 0, 2 : 3, 3 : 2), \{\{A, 2\}, \{B, 1\}, \{C, 3\}\})$, and we are done.

We found the market equilibrium to be $(\vec{p}, M) = ((1 : 0, 2 : 3, 3 : 2), \{\{A, 2\}, \{B, 1\}, \{C, 3\}\})$. The maximum social value is therefore $v(A, 2) + v(B, 1) + v(C, 3) = 12 + 7 + 10 = 29$.

Our algorithm found exactly this market equilibrium.

2 Question 8

(b) In this part we analyze how the prices output by the VCG mechanism compare with the ones output by the algorithm of Theorem 8.8 (finding a market equilibrium (p, M)). The following are the examples we analyze and their corresponding results for each mechanism:

1. Example 1:

Values for b_1 : $\{i_1 : 3, i_2 : 0, i_3 : 15\}$

b_1

i_1

Values for b_2 : $\{i_1 : 11, i_2 : 12, i_3 : 8\}$

b_2

i_2

Values for b_3 : $\{i_1 : 14, i_2 : 13, i_3 : 18\}$

b_3

i_3

Values for b_4 : $\{i_1 : 3, i_2 : 18, i_3 : 5\}$

b_4

Values for b_5 : $\{i_1 : 15, i_2 : 9, i_3 : 1\}$

b_5

Values for b_6 : $\{i_1 : 14, i_2 : 17, i_3 : 9\}$

b_6

Values for b_7 : $\{i_1 : 14, i_2 : 17, i_3 : 13\}$

b_7

And we observe that the prices output by the VCG mechanism and the algorithm of Theorem 8.8 are the same (the matching is also the same because we used the same algorithm to compute the socially optimal state as part of the VCG mechanism)

2. Example 2:

Values for b_1 : $\{i_1 : 12, i_2 : 14, i_3 : 16, i_4 : 8, i_5 : 6, i_6 : 17\}$

b_1

i_1

Values for b_2 : $\{i_1 : 11, i_2 : 7, i_3 : 9, i_4 : 19, i_5 : 1, i_6 : 11\}$

b_2

i_2

Values for b_3 : $\{i_1 : 18, i_2 : 13, i_3 : 17, i_4 : 17, i_5 : 2, i_6 : 16\}$

b_3

i_3

Values for b_4 : $\{i_1 : 15, i_2 : 0, i_3 : 4, i_4 : 1, i_5 : 15, i_6 : 15\}$

b_4

i_4

Values for b_5 : $\{i_1 : 7, i_2 : 8, i_3 : 5, i_4 : 12, i_5 : 18, i_6 : 13\}$

b_5

i_5

Values for b_6 : $\{i_1 : 7, i_2 : 19, i_3 : 8, i_4 : 12, i_5 : 4, i_6 : 1\}$

b_6

i_6

And we observe that the prices output by the VCG mechanism and the algorithm of Theorem 8.8 are the same (the matching is also the same because we used the same algorithm to compute the socially optimal state as part of the VCG mechanism)

3. Example 3:

Values for b_1 : $\{i_1 : 8, i_2 : 11, i_3 : 0, i_4 : 3, i_5 : 6, i_6 : 7\}$

b_1

i_1

Values for b_2 : $\{i_1 : 19, i_2 : 14, i_3 : 15, i_4 : 14, i_5 : 14, i_6 : 16\}$

b_2

i_2

Values for b_3 : $\{i_1 : 17, i_2 : 19, i_3 : 19, i_4 : 13, i_5 : 8, i_6 : 17\}$

b_3

i_3

Values for b_4 : $\{i_1 : 2, i_2 : 15, i_3 : 1, i_4 : 18, i_5 : 11, i_6 : 10\}$

b_4

i_4

Values for b_5 : $\{i_1 : 8, i_2 : 9, i_3 : 7, i_4 : 15, i_5 : 6, i_6 : 10\}$

b_5

i_5

Values for b_6 : $\{i_1 : 12, i_2 : 15, i_3 : 15, i_4 : 8, i_5 : 2, i_6 : 1\}$

b_6

i_6

And we observe that the prices output by the VCG mechanism and the algorithm of Theorem 8.8 are the same (the matching is also the same because we used the same algorithm to compute the socially optimal state as part of the VCG mechanism)

4. Example 4:

Values for b_1 : $\{i_1 : 5, i_2 : 3, i_3 : 0, i_4 : 7, i_5 : 10, i_6 : 5, i_7 : 17, i_8 : 6, i_9 : 18, i_{10} : 8\}$

b_1

i_1

Values for b_2 : $\{i_1 : 5, i_2 : 4, i_3 : 6, i_4 : 9, i_5 : 15, i_6 : 9, i_7 : 17, i_8 : 2, i_9 : 10, i_{10} : 14\}$

b_2

i_2

Values for b_3 : $\{i_1 : 10, i_2 : 11, i_3 : 10, i_4 : 6, i_5 : 4, i_6 : 10, i_7 : 16, i_8 : 11, i_9 : 10, i_{10} : 6\}$

b_3

i_3

Values for b_4 : $\{i_1 : 2, i_2 : 19, i_3 : 4, i_4 : 12, i_5 : 5, i_6 : 8, i_7 : 12, i_8 : 0, i_9 : 11, i_{10} : 11\}$

b_4

i_4

Values for b_5 : $\{i_1 : 18, i_2 : 7, i_3 : 15, i_4 : 11, i_5 : 7, i_6 : 4, i_7 : 2, i_8 : 9, i_9 : 9, i_{10} : 8\}$

b_5

i_5

Values for b_6 : $\{i_1 : 5, i_2 : 2, i_3 : 2, i_4 : 5, i_5 : 1, i_6 : 12, i_7 : 13, i_8 : 18, i_9 : 8, i_{10} : 1\}$

b_6

i_6

i_7

i_8

i_9

i_{10}

And we observe that the prices output by the VCG mechanism and the algorithm of Theorem 8.8 are the same (the matching is also the same because we used the same algorithm to compute the socially optimal state as part of the VCG mechanism)

5. Example 5:

Values for b_1 : $\{i_1 : 15, i_2 : 3, i_3 : 1, i_4 : 5\}$	b_1	i_1
Values for b_2 : $\{i_1 : 11, i_2 : 11, i_3 : 16, i_4 : 5\}$	b_2	i_2
Values for b_3 : $\{i_1 : 9, i_2 : 15, i_3 : 13, i_4 : 17\}$	b_3	i_3
Values for b_4 : $\{i_1 : 15, i_2 : 11, i_3 : 10, i_4 : 16\}$	b_4	i_4
Values for b_5 : $\{i_1 : 19, i_2 : 0, i_3 : 12, i_4 : 7\}$	b_5	
Values for b_6 : $\{i_1 : 17, i_2 : 16, i_3 : 13, i_4 : 9\}$	b_6	

And we observe that the prices output by the VCG mechanism and the algorithm of Theorem 8.8 are the same (the matching is also the same because we used the same algorithm to compute the socially optimal state as part of the VCG mechanism)

That is, in all examples we analyzed, the prices output by the VCG mechanism and the algorithm of Theorem 8.8 were the same, and the matching was also the same because we used the same algorithm to compute the socially optimal state as part of the VCG mechanism. We analyzed far more examples besides the ones presented here, and the results were consistent across all of them—the prices output by the VCG mechanism and the algorithm of Theorem 8.8 were the same (and the matching was also the same because we used the same algorithm to compute the socially optimal state as part of the VCG mechanism).

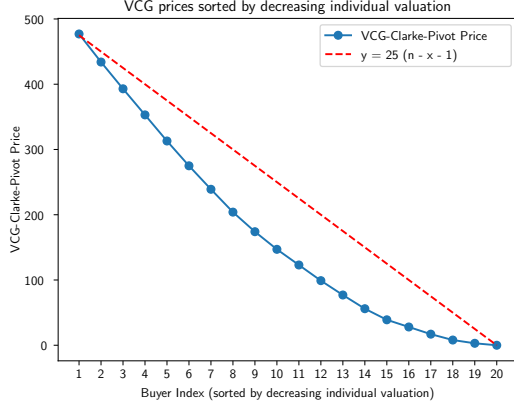
We found evidence online that this phenomenon is not a coincidence, detailed further in [1].

3 Bonus Question 2

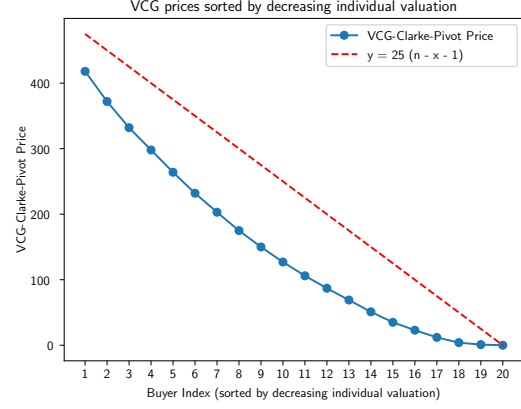
- (a) We structure a *markets-for-bundles* context of identical goods as a simple matching market context, where each bundle B_j 's value for bidder b_i is the product of the value of b_i for the good and the amount of goods in the bundle, c_i . That is, $v_i(B_j) = c_j \cdot t_i$, where t_i is the value of b_i for the singular good, and c_j is the amount of goods in bundle B_j . We then run the VCG algorithm we implemented in the previous part on a few randomized examples of such a markets-for-bundles context, where there are $n = m = 20$ bundles and bidders, and the values t_i are randomized between 1 and 50, and where $c_j = j$ ($j \in \{1, 2, \dots, 20\}$).

Figure 1 summarizes the results of the VCG algorithm on 4 such randomized examples of markets-for-bundles contexts (the results remained the same for other examples we ran). The x -axis represents the index of the bidder, where the bidders are sorted by decreasing order of their individual valuation t_i for the singular good, and the y -axis represents the VCG price for the bidder—commonly referred to as the *externalities* of the bidder on the market.

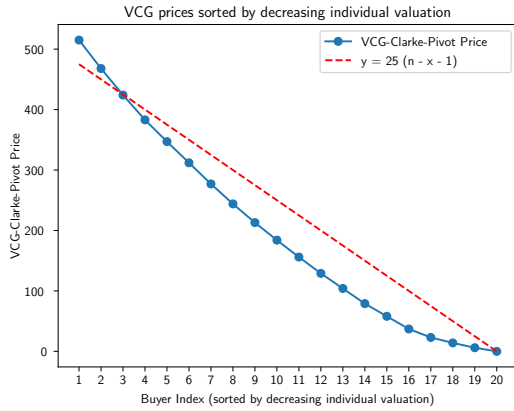
We observe a clear trend, where the VCG prices—i.e. *externalities* of bidders, are decreasing as the valuation of the bidders for the singular good (post tie-breaks of valuations) decreases, which is expected given the structure of the markets-for-bundles context.



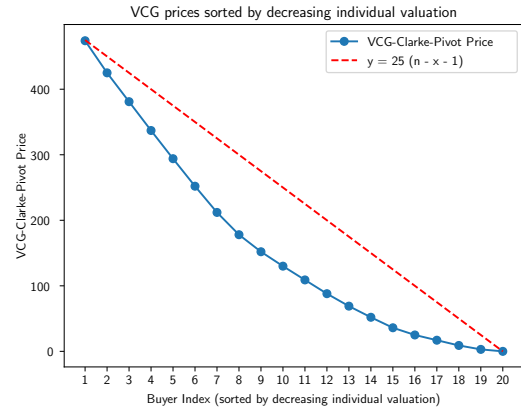
(a) Example 1



(b) Example 2



(c) Example 3



(d) Example 4

Figure 1: VCG prices for bidders in markets-for-bundles contexts

As we've seen in class, the socially optimal assignment in a *markets-for-bundles* of identical goods, is that the larger bundles are assigned to the higher valuation of a singular good. Thus, as the valuation of the singular good increases, the bidder b_i (w.l.o.g. the bidders are sorted by decreasing valuation of the singular goods) is assigned a larger bundle B_i (w.l.o.g. the bundles are sorted by decreasing sizes c_i). Without said bidder b_i , the bundles assignment shifts for the lower-bidding bidders. That is, $\forall j > i$, b_j is changed to be assigned B_{j-1} . This sets the externality to be $\sum_{j=i+1}^n (c_{j-1} - c_j) \cdot t_j$ for bidder b_i . This is the price, and in our case, it's $p_i^{VCG} = \sum_{j=i+1}^n t_j$.

We are thus able to conclude that as the valuation of the singular good decrease (i.e., as i grows), the externality of the bidder (of b_i) should decrease. Moreover, due to the uniformity of the randomization of the values $t_i \sim U([a, a + 1, \dots, b])$ in our context (where $a = 1, b = 50$), and as $c_j - c_{j+1} = 1$ in our context, we should see the externalities remaining under the $y = \left(\frac{a+b}{2}\right) \cdot (n-x-1)$ line, and approaching it as the valuation of the bidders for the singular good increases. This is in fact what we see in Figure 1.

- (b) We implemented a GSP pricing mechanism in a *matching-market* context. The following are comparisons of the prices output by the GSP mechanism and the VCG mechanism over different contexts.

Randomized Matching-Market Contexts: We use the randomization scheme described in the previous part to generate randomized matching-market contexts, where there are $n = m = 20$ bidders and goods, and the values t_i are randomized between 1 and 50. We then run the GSP pricing mechanism we implemented on a few such randomized examples of matching-market contexts, and compare the prices output by the GSP mechanism to the VCG prices.

Figure 2 summarizes the results of the GSP pricing mechanism on 4 such randomized examples of matching-market contexts (the results remained the same for other examples we ran), compared to the VCG prices. The x -axis represents the index of the bidder, where the bidders are sorted by decreasing order of their individual valuation t_i for the singular good, and the y -axis represents the GSP price for the bidders in orange and the VCG price for the bidder in blue.

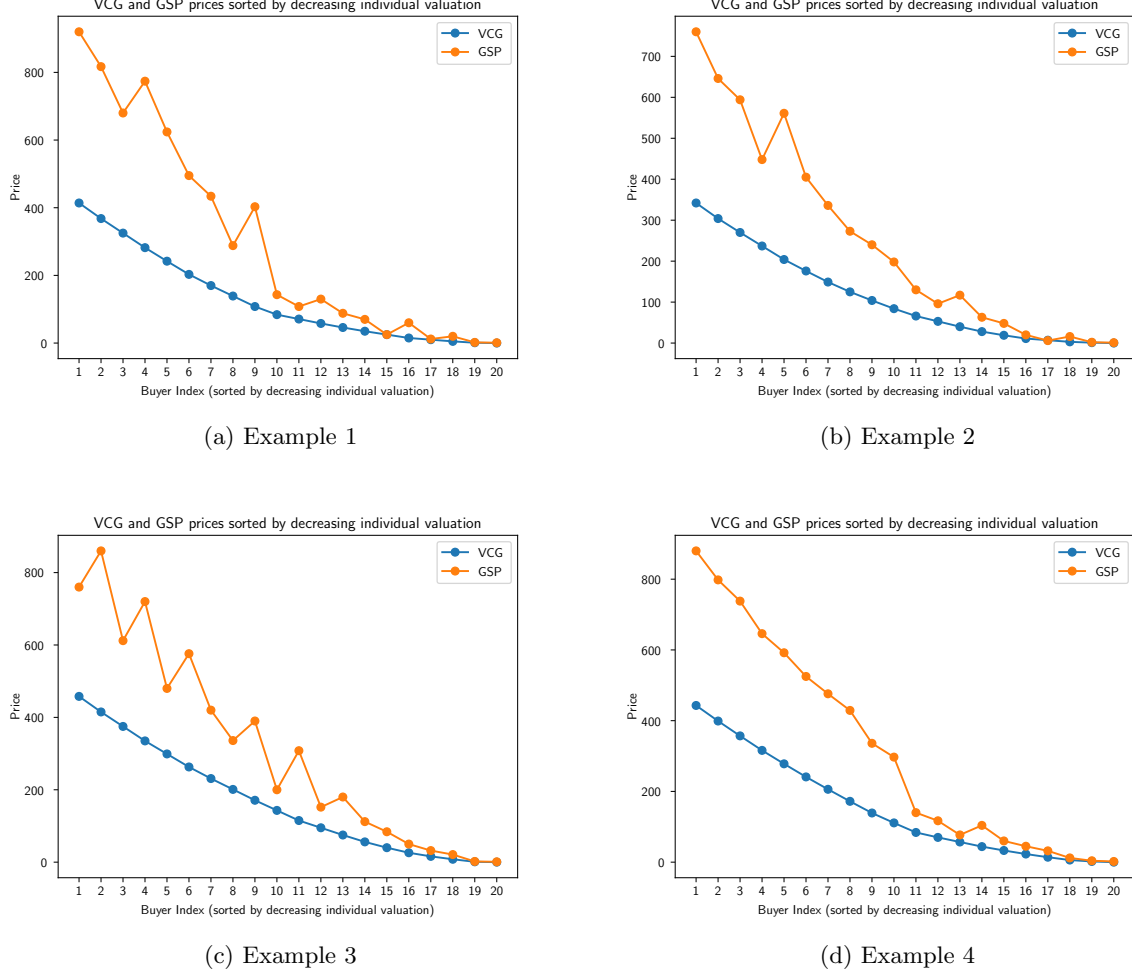


Figure 2: GSP and VCG prices for bidders in matching-market contexts

Unsurprisingly, the GSP prices are also decreasing as the valuation of the bidders for the singular good (up to tie-breaks) decreases. This is expected, as for every bidder b_i , the price/item is set to be the valuation of the next bidder t_{i+1} (where $t_{n+1} = t_n$), and the larger bundles are assigned to the higher-valuing bidders (as in the VCG mechanism). Thus, the GSP price for the bidder, denoted by $p_i^{GSP} = t_{i+1} \cdot c_i$, is decreasing as the valuation of the bidders for the singular good decreases (i.e., as i grows). This is in fact what we see in Figure 2.

In our case, this price is $p_i^{GSP} = t_{i+1} \cdot (n - i + 1) \geq \sum_{j=i+1}^n t_j = p_i^{VCG}$. Thus, we expect that the GSP prices will be greater than the VCG prices, and this is indeed what we see in Figure 2.

Contexts in which VCG and GSP prices are similar As already highlighted, for the VCG mechanism, the externality of a matched bidder b_i —i.e. for $i \in [\min(n, m)]$ (where n, m are the amount of bidders and bundles respectively), assuming bidders and bundles are ordered (w.l.o.g) by decreasing valuation of the singular good and decreasing bundle size respectively, is its price $p_i^{VCG} = \sum_{j=i+1}^n (c_{j-1} - c_j) \cdot t_j$. This is in contrast to the GSP mechanism, where the price is $p_i^{GSP} = t_{i+1} \cdot c_i$.

Therefore, a context in which those are similar prices would be where $\forall j \in [m]. c_j = 1$ (as we've demanded so far) and where the valuations differ very little from each other by some small constant

$a \in \mathbb{N}$, such that $\forall i \in [n]. a \cdot n^2 \ll t_i$. To take an example, we set $n = m = 10$, $a = 1$, and $t_i = 1000 + (n - i)$. In such a context,

$$p_i^{VCG} = \sum_{j=i+1}^n (c_{j-1} - c_j) \cdot t_j = \sum_{j=i+1}^n t_j \approx t_{i+1} \cdot (n - i) - \sum_{j=1}^{n-i-1} a \cdot j \approx t_{i+1} \cdot (n - i) - a \cdot \frac{(n-i-1)(n-i)}{2} \approx t_{i+1} \cdot (n - i) \approx t_{i+1} \cdot c_i = p_i^{GSP}$$

This is indeed what we see in [Figure 3](#).

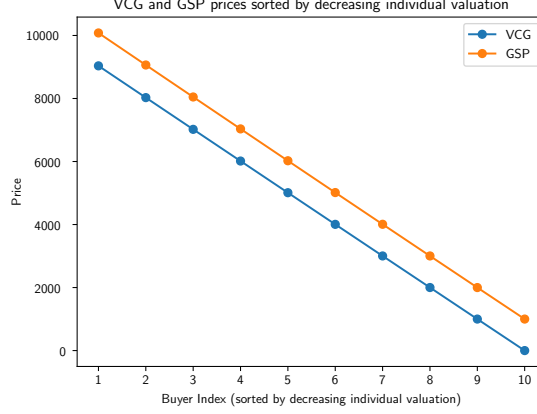


Figure 3: VCG and GSP prices for bidders in a context where they are similar

An additional context in which the VCG and GSP prices are similar is when the valuations of the bidders for the singular good are the same, and the bundle sizes are such that $c_i = (n - i + 1)$. In this case,

$$p_i^{VCG} = \sum_{j=i+1}^n (c_{j-1} - c_j) \cdot t_j = t_{i+1} \cdot (n - i) \approx t_{i+1} \cdot c_i = p_i^{GSP}$$

Contexts in which VCG and GSP prices are wildly different A context in which the VCG and GSP prices are wildly different is where the bundle sizes c_i (w.l.o.g they are ordered by decreasing size), satisfy the following:

$$\begin{aligned} \forall i \in [1, m-1]. c_i &= c_{i+1} + 1 \\ \text{and} \\ \forall i \in [m]. c_i &\gg (n - i + 1) \end{aligned}$$

This induces that

$$p_i^{VCG} = \sum_{j=i+1}^n (c_{j-1} - c_j) \cdot t_j = \sum_{j=i+1}^n t_j \leq t_{i+1} \cdot (n - i) \ll t_{i+1} \cdot c_i = p_i^{GSP}$$

This is indeed what we see in [Figure 4](#), for which $n = m = 10$, $c_i = 100 + (n - i)$ and $t_i = n - i + 1$.

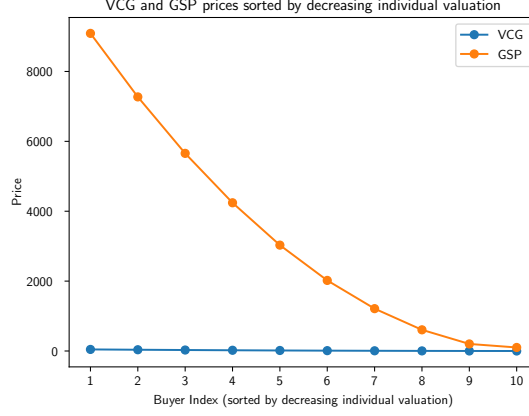


Figure 4: VCG and GSP prices for bidders in a context where they are different

An additional context in which the VCG and GSP prices are wildly different is when the bundle sizes are relatively small and similar, and where the valuations of the bidders for the singular good are such that $t_i \gg t_{i+1} \cdot (n - i)$. For instance, take $n = m = 10$, $t_i = 2^{i+3}$, and $c_i = n - i + 1$. In this case,

$$p_i^{VCG} = \sum_{j=i+1}^n (c_{j-1} - c_j) \cdot t_j = \sum_{j=i+1}^n t_j \ll t_{i+1} \cdot (n - i) \approx t_{i+1} \cdot c_i = p_i^{GSP}$$

This is indeed what we see in Figure 5.

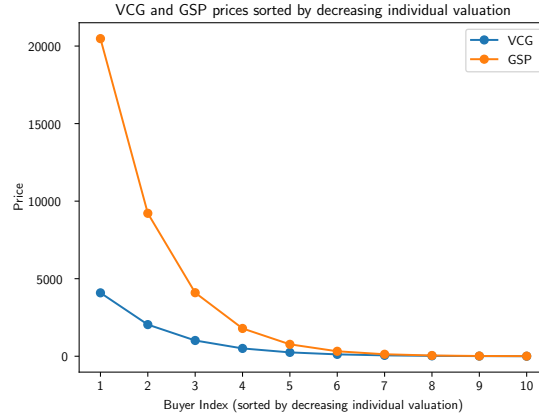


Figure 5: VCG and GSP prices for bidders in a context where they are different

- (c) We turn to analyze the BRD procedure on a GSP matching market auction, here a player's strategy is their valuation report. We have in place mechanisms inside the procedure that identify cycles in order to break them and report a divergence of the procedure.

After implementing the *BRD on GSP matching market auctions* procedure, we gather statistics about the convergence of the BRD procedure on a GSP matching market auction. We run the BRD procedure on many randomized examples such as in part (a), for different configurations of n, m and the maximum valuation t (which was 50 in part (a)). We run ≈ 20 randomized examples from each of the following configurations, all of which had $t = 15$:

- $n = 5 = m$
- $n = 5 \ll 20 = m$
- $m = 5 \ll 20 = n$

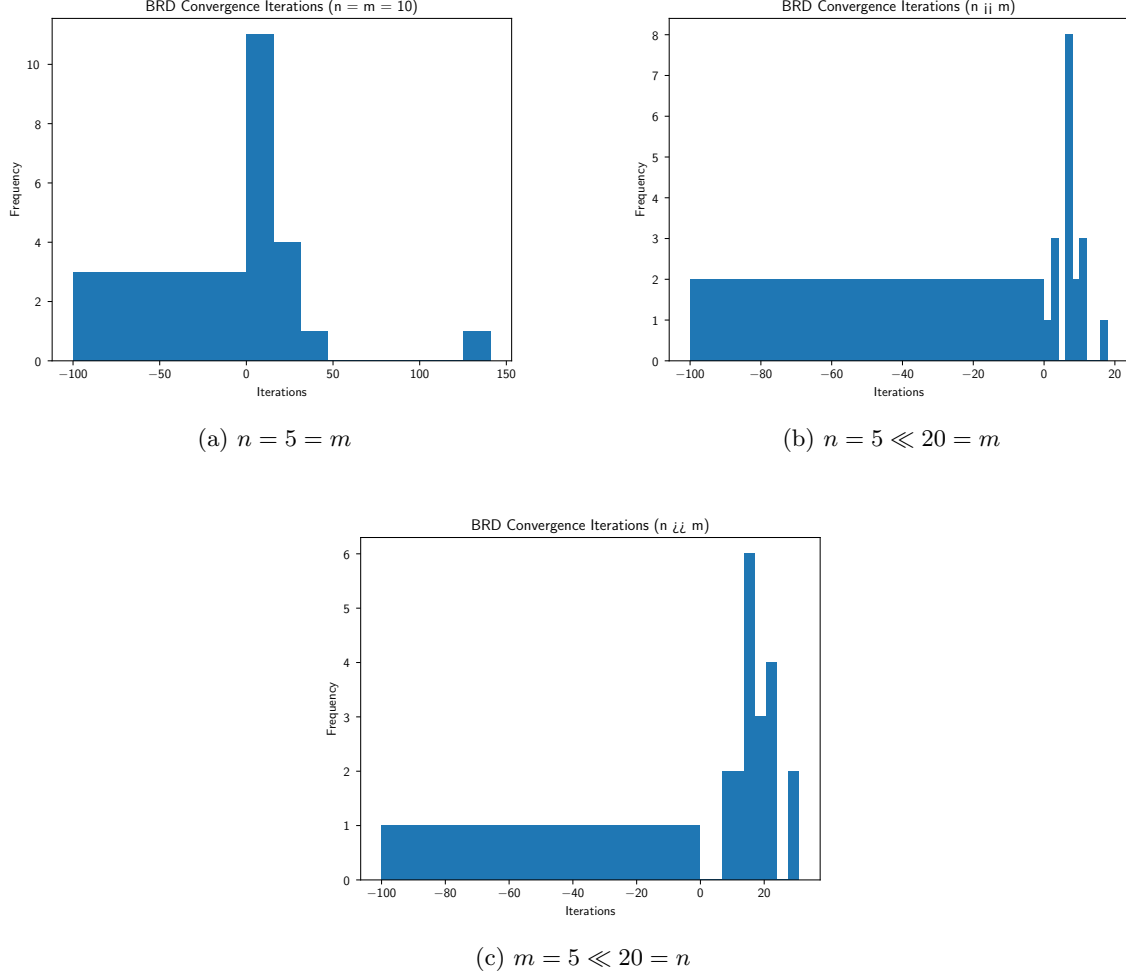
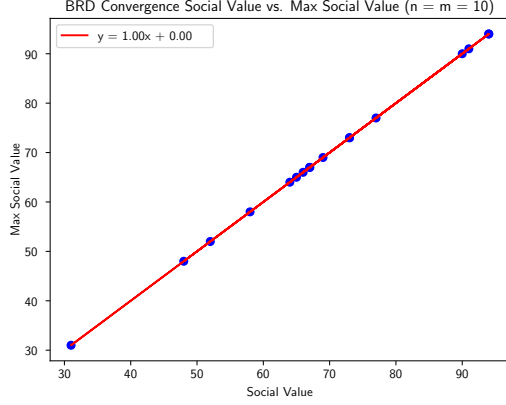


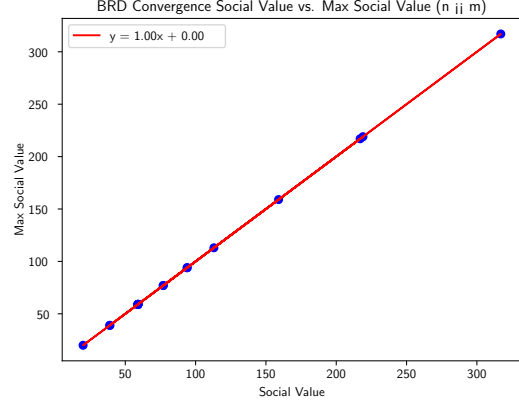
Figure 6: Histograms of the iteration count for the BRD procedure on GSP matching market auctions. Negative values indicate divergence.

The results were consistent among all of them: the BRD procedure converged in all cases around 90% of the time, and for some relatively small amounts of ≈ 20 iterations (we observe what appears to be statistical noise in the iteration measurement for $n = 5 = m$). This is depicted in Figure 6.

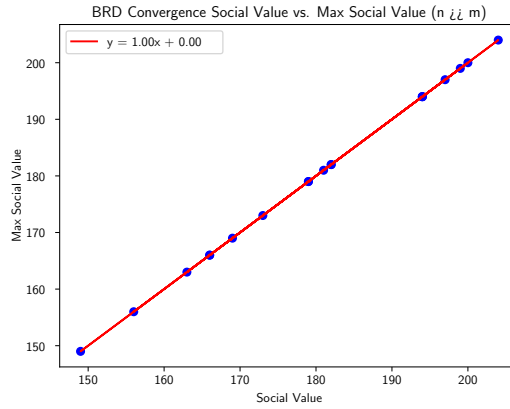
We remind ourselves that the GSP mechanism *Nash-implements social value maximization in any market-for-bundles context*. That is, there exists a report profile $\vec{r}$ such that $GSP^\Gamma(\vec{r})$ for context Γ is the socially optimal outcome, and such that $\vec{r}$ is a Nash equilibrium in the GSP^Γ mechanism. Therefore, we sought to analyze the social value of the outcomes of the BRD procedure on the GSP mechanism when it converges. We found that the social value of the outcomes of the BRD procedure on the GSP mechanism when it converges is indeed the same as the maximum social value in the context. This is depicted in Figure 7.



(a) $n = 5 = m$



(b) $n = 5 \ll 20 = m$



(c) $m = 5 \ll 20 = n$

Figure 7: Scatter plots where each dot corresponds to a different randomized context. The x -axis represents the social value of the outcome of the BRD procedure on the GSP mechanism when it converges on the context, and the y -axis represents the maximum social value in the context. The scatter plot is over the results of the previous BRD runs in Figure 6. The red line represents the line $y = x$.

Part 5: Exchange Networks for Uber

We will construct a simplified market scenario for a ride-sharing app like Uber. Our world will consist of an $\ell \times \ell$ grid, and there will be two types of participants, riders and drivers.

- A rider R is specified by a current location $(x_0, y_0) \in [\ell] \times [\ell]$, a desired destination $(x_1, y_1) \in [\ell] \times [\ell]$, and a value for reaching that destination.
- A driver D is specified by a current location $(x_0, y_0) \in [\ell] \times [\ell]$.

We define the cost of a matching between a rider R and a driver D , $c(R, D)$, to be the distance from the driver to the rider and then to the destination (measured via Manhattan distance, i.e., L^1 distance).

1 Question 9

We are tasked with encoding the above as an exchange network. Namely, defining a graph $G = (V, E)$ where the vertices are the riders and drivers and there is an edge between every rider and driver.

Denoting the value for rider R reaching his destination as v_R , and reminded that the value of an edge in an exchange network must be a natural number, we define the values associated with each edge in the graph in the following way:

$$v(\overrightarrow{RD}) = \begin{cases} v_R - c(R, D) & \text{if } v_R \geq c(R, D) \\ 0 & \text{otherwise} \end{cases}$$

This value function ensures economic rationality by capping the cost of a match by the rider's valuation of reaching their destination—done by setting the value of the edge to zero if the rider's valuation is less than the cost of the match. This ensures that given any stable outcome, we can extrapolate a stable outcome where non-beneficial matches (i.e., where the value is realistically negative) are removed—where stability is maintained as the allocation of such rider-driver pairs would be 0 pre-removal (as the edge's value between them is 0) and 0 post-removal (as unmatched nodes have a value of 0).

2 Question 10

In the following we analyze the prices and profits in the computed stable outcomes for the exchange networks we defined in the previous part. The prices and profits are derived from the computed stable outcome (M, a) . For a rider R_i and driver D_j match in the stable outcome, the rider price is

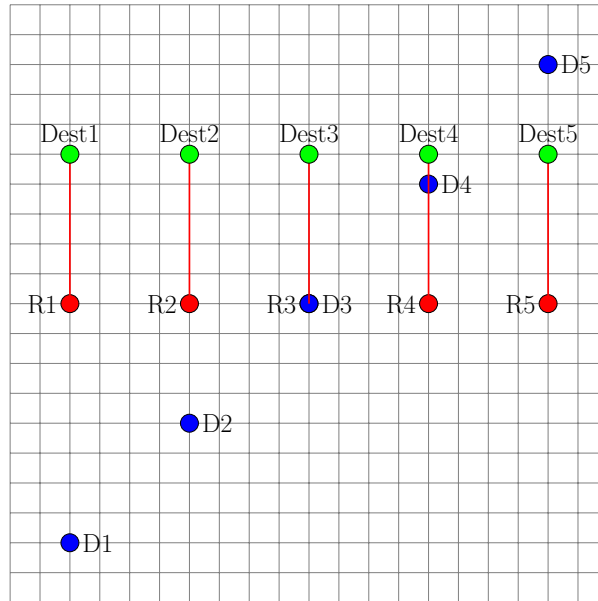
$$v_{R_i} - a(R_i) = c(R_i, D_j) + ((v_{R_i} - c(R_i, D_j)) - a(R_i)) = c(R_i, D_j) + (v(R_i, D_j) - a(R_i)) = c(R_i, D_j) + a(D_j)$$

Where $v(R_i, D_j)$ denotes the value of the edge between R_i and D_j in the exchange network, which per our definition is $v_{R_i} - c(R_i, D_j)$. Conversely, the driver profit is

$$(v_{R_i} - a(R_i)) - c(R_i, D_j) = (v_{R_i} - c(R_i, D_j)) - a(R_i) = v(D_j, R_i) - a(R_i) = a(D_j)$$

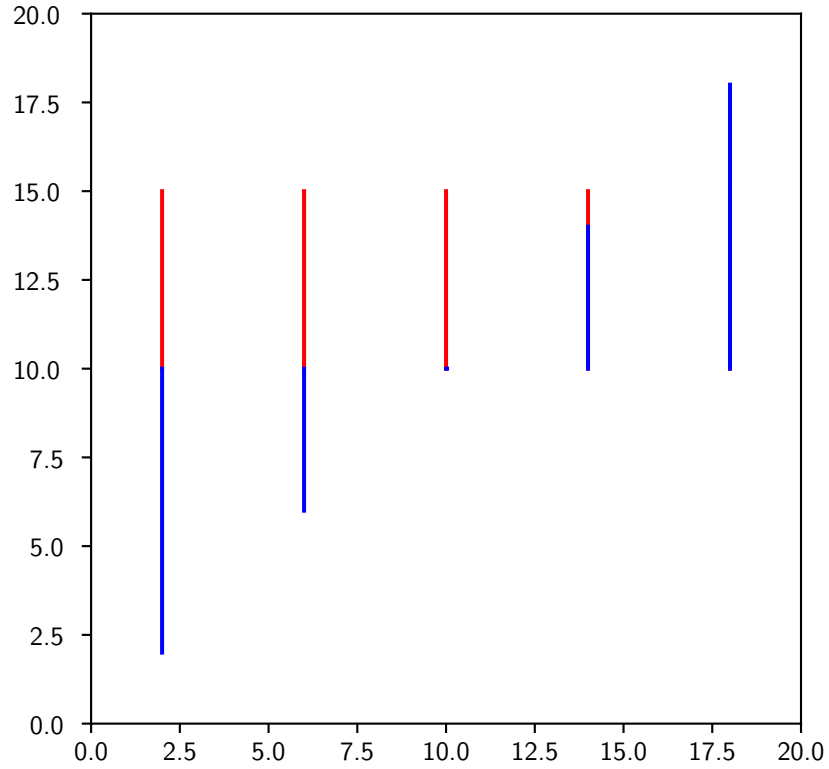
- (a) We consider the following examples of exchange networks for Uber-like ride-sharing apps, show their computed stable outcome, and argue the legitimacy of the computed driver profits and rider prices.

Example 1 Depicted by the following grid representation, where red circles represent riders and blue circles represent drivers (note that the center of the grid has a rider and a driver, yet only the driver is shown), and where red lines from the rider location (red circle) to the destination (green circle) represent the path the rider would like to take:



Where the rider values are all set to 30.

The matching of the stable outcome illustrated by connecting a driver to the rider by a blue line, and by connecting the rider's location to the destination by a red line, for each edge between a rider and a driver:



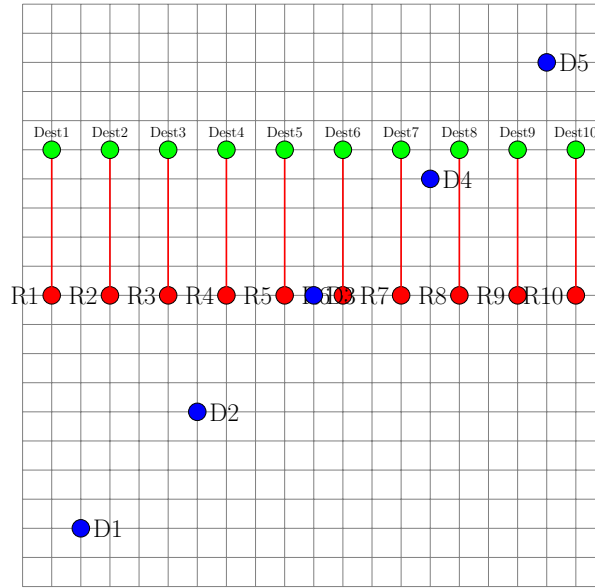
And the driver profits and rider prices for this stable outcome are as follows:

- Driver Profits: $[0, 0, 0, 0, 0]$
- Rider Prices: $[13, 9, 5, 9, 13]$

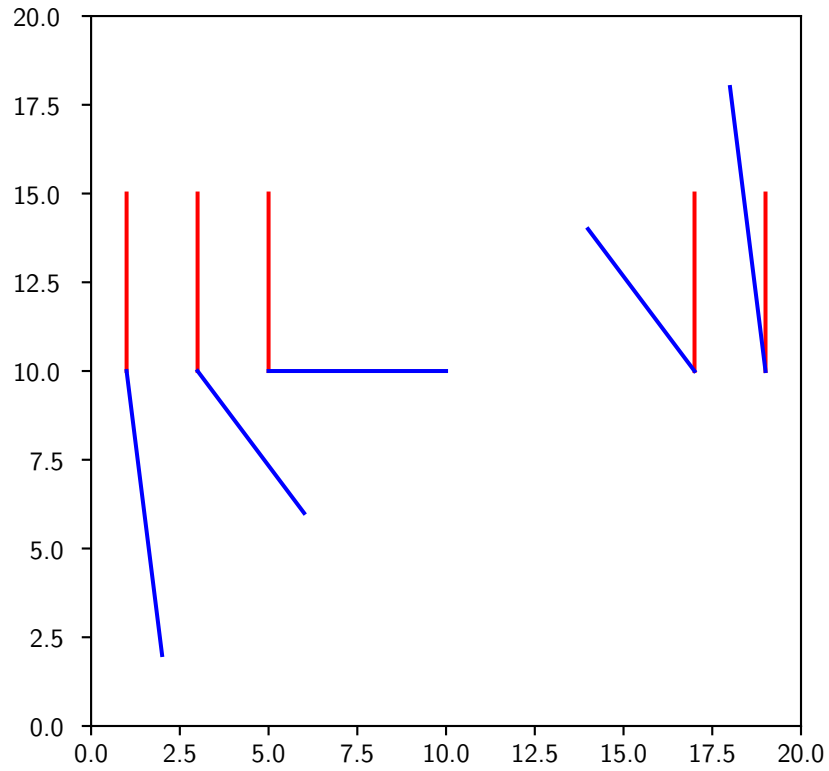
We observe driver profits all to be 0. This is sensible, as our algorithm from Question 9, which encodes the exchange network as a matching market context, encodes the riders being the buyers and the drivers being the sellers. As the stable outcome is calculated by the market equilibrium algorithm which prioritizes lower prices for the buyers, the profits are thus expected to be on the lower end.

Rider prices, on the other hand, depict that network position greatly affects the ability to extract more value due to more opportunities/alternatives. As the rider is closer to the center, its average distance from all drivers is lower (where its distance from its destination remains the same), which heightens competition between drivers for the rider, and thus the rider can extract more value from the drivers.

Example 2 Depicted by the following grid representation is the third example, using similar conventions as the first example:



The matching of the stable outcome illustrated by connecting a driver to the rider by a blue line, and by connecting the rider's location to the destination by a red line, for each edge between a rider and a driver:



And the driver profits and rider prices for this stable outcome are as follows:

- Driver Profits: [25, 27, 27, 25, 23]
- Rider Prices: [39, 39, 37, 35, 30, 25, 30, 35, 37, 37]

We observe that driver profits increase—which is expected given that the demand outgrew the supply, which thus heightened the competitive overbidding nature among the riders, amplified by the stability property of the outcome. Rider prices also increase, which is expected as the riders are competing with each other to provide the highest price to the drivers, and thus higher prices have increasing frequency.

Yet again, we see that riders positioned closer to the center of the grid can extract more value from the drivers, as their average distance from all drivers is lower (where their distance from their destination remains the same), which heightens competition between drivers for the rider, and thus the rider can extract lower prices. Similarly, drivers positioned closer to the center of the grid can extract more value from the riders, as their average distance from all riders is lower, which heightens competition between riders for the driver, and thus the driver can extract higher profits.

- (b) We implemented a procedure that on a 100×100 grid, generates an exchange network context of n riders and m drivers with random locations and destinations, where each rider has value 100, and computes the stable outcome for this context. The following is the analysis of the results when running over different configurations of (n, m) for many times. We consider the prices and profits for different configurations of (n, m) —each exhibiting a different relationship between n and m . We show the distribution of the prices and profits for the riders and drivers in each configuration, utilizing a histogram to depict the *probability density function* of the prices/profits, with additional statistics such as the min, max, mean, median, standard deviation, and the amount of times the rider/driver was matched.

Configuration 1: $n = m = 10$ This configuration exhibits a scenario where there is an equal number of drivers and riders. The following is the analysis of the distribution of the prices/profits for each individual rider/driver in this configuration, after using our procedure to generate random instances of this configuration:

- **Driver Profits:** We identify one prominent distributions of the driver profits in this configuration, depicted in Figure 8. The distribution is a right-skewed distribution, which accommodates most of the profits at the lower end (0-7) of the spectrum (80%), with the rest being spread across the higher end of the spectrum (7-30), with gradually decreasing frequency as the profits increase.

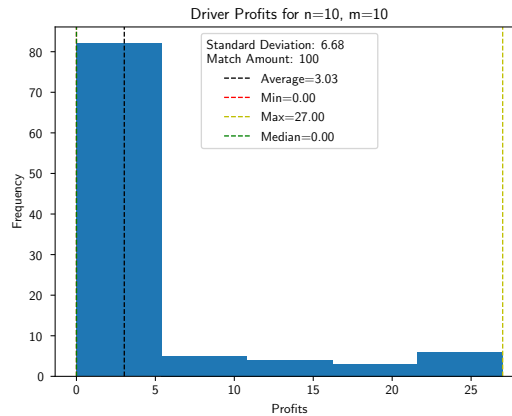


Figure 8: Driver Profits Distribution for $n = m = 10$

The former is expected due to our algorithm from Question 9, which encodes the exchange network as a matching market context where the riders being the buyers and the drivers being the sellers. As the stable outcome is calculated by the market equilibrium algorithm which prioritizes lower prices for the buyers, the profits are thus lower.

The latter is also expected as the higher profits are, the more likely they are undercut by the other drivers—i.e., the drivers are competing with each other to provide the lowest price to the riders, and thus higher profits have decreasing frequency.

- **Rider Prices:** We identify one prominent distributions of the rider prices in this configuration, depicted in Figure 9. The distribution is a left-skewed distribution, which favors higher prices, with most of the prices being at the higher end of the spectrum (85-100), with gradually decreasing frequency as the prices decrease.

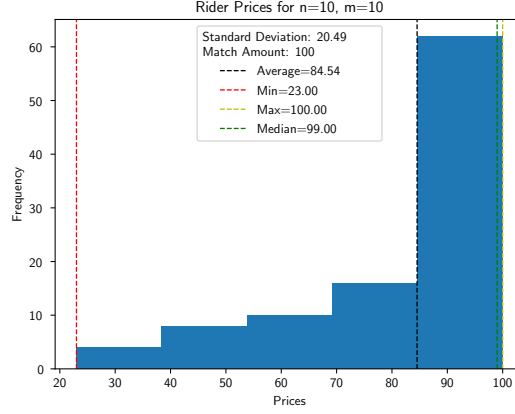


Figure 9: Rider Prices Distribution for $n = m = 10$

It is noticeable that the density of the prices is more spread out than the driver's profits. As already mentioned above, this is expected as our algorithm's nature is to prioritize lower prices for the buyers.

As for the decreasing frequency as the prices decrease, this is also expected, as the lower the price, the more likely it is overbid by other riders—i.e., the stability requirement of the outcome induces that the riders compete with each other to provide the highest price to the drivers, and thus lower prices have decreasing frequency.

Configuration 2: $5 = n \ll m = 20$ This configuration exhibits a scenario where there are more drivers than riders—an example of higher supply than demand. The following is the analysis of the distribution of the prices/profits for each individual rider/driver in this configuration, after using our procedure to generate random instances of this configuration:

- **Driver Profits:** We identify one prominent distributions of the driver profits in this configuration, depicted in Figure 10. The distribution is that of having virtually all profits at 0, with very few times where the profits are non-zero. Naturally we also see that match count for drivers drop significantly from 100 (in the $n = m = 10$ configuration) to ≈ 40 .

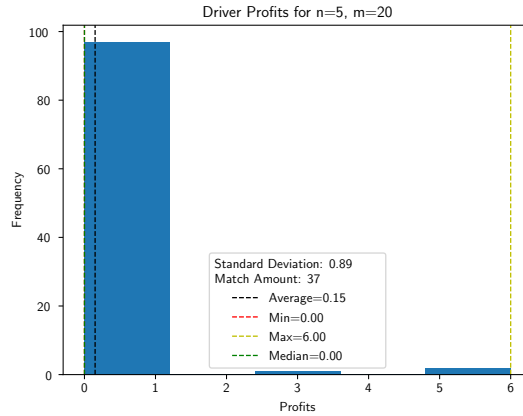


Figure 10: Driver Profits Distribution for $5 = n \ll m = 20$

The distribution is expected, as the more drivers than riders, the more supply than demand, and thus the more likely it is that some drivers will not be matched with any riders, and thus have a profit of 0. This also heightens the competitiveness between drivers described in the previous configuration. This explains why we see such a drop in the average profit per driver (from ≈ 3 in the $n = m = 10$ configuration to ≈ 0 in this configuration).

- **Rider Prices:** We identify one prominent distributions of the rider prices in this configuration, depicted in Figure 11. The distribution is a more spread-out version of the rider prices in the $n = m = 10$ configuration, with the prices being more spread out across the spectrum, with much higher frequency at the lower ends of the spectrum and a significant drop in average rider price (from ≈ 83 in the $n = m = 10$ configuration to ≈ 70 in this configuration).

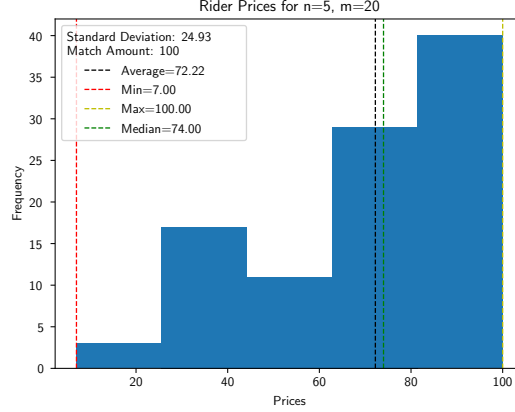


Figure 11: Rider Prices Distribution for $5 = n \ll m = 20$

This is expected, as the higher supply than demand means that the riders have more options to choose from, and thus the prices have more density on the lower end in this configuration (due to heightened competition between drivers), compared to the previous configuration.

Configuration 3: $20 = n \gg m = 5$ This configuration exhibits a scenario where there are more riders than drivers—an example of higher demand than supply. The following is the analysis of the distribution of the prices/profits for each individual rider/driver in this configuration, after using our procedure to generate random instances of this configuration:

- **Driver Profits:** We identify one prominent distributions of the driver profits in this configuration, depicted in Figure 12. The distribution is structured similarly to the previous configurations, only that there is much more density at the higher ends of the spectrum, with the average profit rising significantly to ≈ 20 , and the match amount rises back to ≈ 100 .

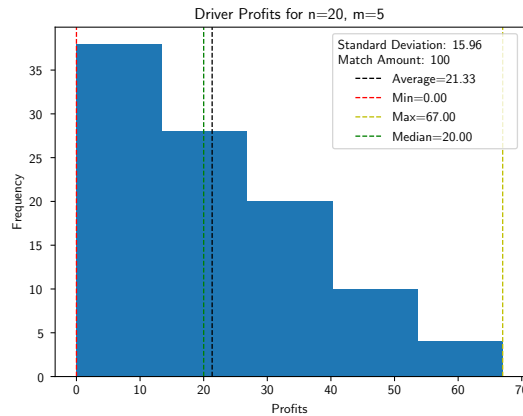


Figure 12: Driver Profits Distribution for $20 = n \gg m = 5$

The distribution is expected, as the more riders than drivers, the more demand than supply, and thus the more likely it is that drivers will be matched with riders, and the more the overbid competition between riders is heightened.

- **Rider Prices:** We identify one prominent distributions of the rider prices in this configuration, depicted in Figure 13. The distribution is a less-spread out version of the rider prices than in the $n = m = 10$ configuration, with the prices being more concentrated at the higher ends of the spectrum, with much higher frequency at the higher ends of the spectrum and a significant rise in average rider price (from ≈ 83 in the $n = m = 10$ configuration to ≈ 96 in this configuration). Additionally, naturally, we observe that the match amount drops significantly from ≈ 100 (in the previous configurations) to ≈ 25 in this configuration.

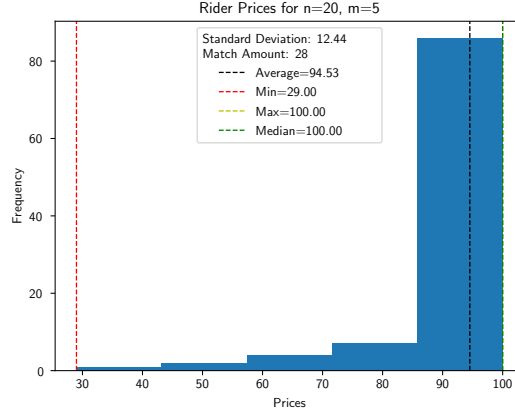


Figure 13: Rider Prices Distribution for $20 = n \gg m = 5$

This is expected, as the higher demand than supply heightens the already-competitive over-bidding nature of the riders under a stable outcome, and thus the prices are more concentrated at the higher ends of the spectrum.

3 Question 11

Denote the original value function of the edge between the rider R and the driver D as $v(\overrightarrow{RD})$ —as defined in Question 9. To account for driver's drive location preferences in our exchange network, we can modify the value function of the edge between a rider R and a driver D to incorporate the driver's preference for their drive location.

This can be done in many ways, for example by lowering the edge value with the Manhattan distance between the rider's location and the closest high-value location to the driver's location. That is

$$\begin{aligned} \hat{v}(\overrightarrow{RD}) &= v(\overrightarrow{RD}) - \min_{HVL \in \text{High-Value Locations}} (\|R's \text{ location} - HVL\|_1) \\ v'(\overrightarrow{RD}) &= \begin{cases} \hat{v}(\overrightarrow{RD}) & \text{if } \hat{v}(\overrightarrow{RD}) \geq 0 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Where $v'(\overrightarrow{RD})$ is the new value function of the edge between the rider R and the driver D .

Using the approach above, we can even incorporate the level of preference the driver has for their drive location by dividing the aforementioned Manhattan distance by the preference level of the high-value location. That is

$$\begin{aligned} \hat{v}(\overrightarrow{RD}) &= v(\overrightarrow{RD}) - \min_{HVL \in \text{High-Value Locations}} \left(\frac{\|R's \text{ location} - HVL\|_1}{D's \text{ preference value of HVL}} \right) \\ v'(\overrightarrow{RD}) &= \begin{cases} \hat{v}(\overrightarrow{RD}) & \text{if } \hat{v}(\overrightarrow{RD}) \geq 0 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Where $v'(\overrightarrow{RD})$ is the new value function of the edge between the rider R and the driver D , and where we assume that the driver has some preference value for each high-value location it acknowledges.

4 Bonus Question 3

We assume that the city implements a public transportation system, where the cost to take the public transportation system is $a + b \cdot \text{dist}(\text{initial location}, \text{destination})$, where a is a fixed base fare and b is some constant factor multiplier. Furthermore, anyone can take this public transportation system from any location.

- (a) To implement a procedure that includes the public transportation option in when computing the stable outcome of transportation network defined in previous questions, we simulated this as n additional drivers emulating public transportation, where the cost of driving is as described, given that there are n riders. We then computed the stable outcome for this new network.

This ensures that all players have the ability to choose public transportation instead of Uber.

Since we still have a bipartite graph, we know that a stable matching must exist, per Theorem 9.4 in the notes.

- (b) We are tasked with choosing different values for a and b and discuss how it affects the prices charged to riders/profits of drivers. For that we note the following:
- Higher values of a mean that public transportation becomes less attractive for everyone.
 - Higher values of b mean that public transportation is more reliable for shorter distances.
 - If $b > 1$, the cost of merely *driving the rider from its location to its destination* with public transportation is higher than with Uber (so with a short distance between an Uber and a rider, it might be preferable to take an Uber, depending on the base fare a).

References

- [1] David Easley and Jon Kleinberg. Chapter 15, section 15.9 of: Networks, crowds, and markets: Reasoning about a highly connected world. pages 26–42. Available at: <https://www.cs.cornell.edu/home/kleinber/networks-book/networks-book-ch15.pdf>.